

# Bogdan Neterebskii

- Senior Software Engineer with 12 years of experience, primarily in backend development, but also possessing expertise in frontend and ML technologies.
- I also bring 3 years of experience leading a team of approximately 25 developers, emphasising transparency and efficiency in communication. I am comfortable in both leadership and developer roles, and committed to remaining actively involved in development projects.
- Highly motivated by constant learning, my focus is on solving business problems.
- I am passionate about research tasks, robotics, 3D printing and AI-related topics.

## Contacts

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## Principles

These principles are my global approach and stay with me across roles, teams, and companies.

- **Do something valuable.** Either make people's lives easier and healthier, or reduce environmental impact. No gambling, high-frequency trading or anything else that feels to me wrong.
- **Transparency first.** [C4 models](#) and [Mermaid](#) diagrams are must-haves for a clear system overview, with Confluence pages, solid READMEs, and how-to blocks maintaining shared context.
- **Focus on the result.** Keep outcomes visible, measure impact, and change course when the value is not there.
- **Default tech stack mindset.** Start with proven tools and borrowed best practices, yet stay open to new solutions when they bring measurable value.
- **Respect time.** Prefer deeper thinking and written pre-mortems over extra meetings; flag risks before large discussions.
- **Responsible AI usage.** Let AI handle routine work, but review every line, understand its limits, and step in manually whenever needed.

"My goal is to make the team more efficient and the work smoother when I'm around than when I'm not."

## Experience

### Senior software developer

📍 [AI13](#) / Belgrade, Serbia

🕒 05.2024 – current time

🔧 Python, React, ThreeJS, Rhino + Grasshopper, RFEM6, Keycloak

I worked on developing services and algorithms to speed up the building construction process, starting from the project brief, through exploring structural options, and finishing with prefabrication in the factory. This involved studying the physics of load-bearing structures and using [FEM](#) analysis tools like [RFEM](#) to ensure safety and optimize structural performance.

I also focused on making workflows more efficient and integrating designs with prefabrication needs to ensure smooth transitions between design and factory production.

## Software developer

📍 [Nortal](#) / *Belgrade, Serbia*

🕒 08.2022 – 05.2024

🔧 Python, Spark, React, Kubernetes, Kafka, Keycloak

I contributed to the development of the digital core for NEOM city. I collaborated with other developers to create a microservice system, ensuring that the architecture diagram was always current and facilitating knowledge sharing within the team. This collaboration was crucial in maintaining a cohesive and efficient development process.

Furthermore, I developed backend systems and assisted in mobile development for various applications designed for NEOM Airport. These applications were developed in a very short period of time and successfully released.

## Senior software developer

📍 [The Anylogic Company](#) / *Saint-Petersburg, Russia*

🕒 09.2021 – 08.2022 / 11 months

🔧 Java, Python, Docker, Gitlab, NSIS

Played a key role in redesigning a standalone application for all OS platforms with a 20-year history of development.

Fixed a bottleneck in the development and release pipeline by setting new a new IaC infrastructure and build pipeline, based on Gitlab solution.

## Head of bioinformatics software development group

📍 [BIOCAD](#) / *Saint-Petersburg, Russia*

🕒 02.2019 – 08.2021 / 2 years 7 months

🔧 Haskell, Python, React, RabbitMQ, Gitlab, Kubernetes

Our team developed two dedicated services for storing and connecting diverse biological data generated during drug discovery, including antibodies, plasmids, viruses, clustering results, sequence annotations, and so on.

We also have developed a couple of dozen algorithms for labs that simplify and speed up the process from computer-based sequences to physical proteins or viruses. Additionally, we created a lab-friendly service with a React and [NextJS](#) frontend, incorporating Haskell and Python algorithms that operate across multiple servers through the Kubernetes queue and API.

## Lead software developer

👉 [BIOCAD](#) / Saint-Petersburg, Russia

🕒 01.2018 – 01.2019 / 1 year 1 month

🔧 Haskell, Python, Neo4j, Mongo

Developed an algorithm to find valid details in other 3D structures for inspiration in rational [in silico](#) modeling, utilizing the entire [PDB](#) database. A poster with this algorithm was presented at [PEGS 2019](#).

Collaborated with the Matrosov Institute for System Dynamics and Control Theory to create optimization methods, enabling faster identification of local extremum in protein structures compared to standard approaches. Published the article "Algorithms for local minimization of 3D molecules OPLS force field" (see below in the articles).

Led a team in creating a project management service using MongoDB and Haskell on the backend, along with React on the frontend, to streamline projects coordination.

## Senior software developer

👉 [BIOCAD](#) / Saint-Petersburg, Russia

🕒 01.2017 – 12.2017 / 1 year

🔧 Haskell, Python, ZeroMQ, Neo4j

Together with my team, we created an algorithm for the [protein folding problem](#). The quality of the resulting 3D antibody structure prediction algorithm is comparable to [Rosetta Software](#) and [Schrödinger Software](#).

Created a message bus for efficient task distribution based on ZeroMQ with bindings for Haskell and Python.

## Software developer

👉 [BIOCAD](#) / Saint-Petersburg, Russia

🕒 05.2016 – 12.2016 / 8 months

🔧 Scala, Julia, MySQL, Docker

Wrote in Scala:

- backend for a service for storing and connecting various biological items

- backend for a proprietary authorization and authentication system that was only changed to [Keycloak](#) and [Azure](#) in 2019
- [Sun Grid Engine](#)-based solution for high-load server

Wrote in Julia:

- [basecalling](#) algorithm for converting raw data into DNA sequence

## Junior software developer

📍 [Laser Systems](#) / *Saint-Petersburg, Russia*

🕒 07.2015 – 04.2016 / *10 months*

🔧 Qt

Developed a position algorithms for laser tools and created a module for connecting various components of the product via protobuf protocol.

## Mathematician

📍 [Kotlin-Novator](#) / *Saint-Petersburg, Russia*

🕒 01.2015 – 06.2015 / *6 months*

🔧 Python, LaTeX

Worked on solving aircraft navigation problems and developed algorithms for trajectory optimization.

## Junior web-developer

📍 Mr.Brooks Private Marketing / *Saint-Petersburg, Russia*

🕒 12.2013 – 12.2014 / *1 year*

🔧 WordPress, JavaScript, CSS, Adobe Illustrator

Created websites using WordPress CMS and customized them with JavaScript. Additionally, I collaborated with designers to create a new font.

## Education

### Saint Petersburg State University, Russia

🕒 09.2010 – 06.2015

Mathematics and Mechanics Faculty

Diploma of Specialist in Mathematical Physics

## Saint Petersburg State University, Russia

🕒 09.2011 – 08.2014

Faculty of Military Studies

Lieutenant

## Diplomas and certificates

- 2009, ICYS, Silver Medal in Mathematics, Pszczyna, Poland
- 2010, ICYS, Gold Medal in Mathematics, Bali, Indonesia
- 2010, Intel ISEF, Third place in Mathematics, San Jose, USA
- 2021, [Deep learning specialization on Coursera](#) ([certificate](#)):
  - [Neural Networks and Deep Learning](#)
  - [Improving Deep Neural Networks: Hyperparameter Tuning, Regularization and Optimization](#)
  - [Structuring Machine Learning Projects](#)
  - [Convolutional Neural Networks](#)
  - [Sequence Models](#)
- 2022, [Modern Robotics: Mechanics, Planning, and Control Specialization](#)
  - [Foundations of Robot Motion](#)

## Presentations

02.2019 [Think like a graph / Думай как граф](#) (in Russian)

Presentation from *FProg SPb* meeting about Neo4j database and Haskell libraries to work with it.

## Articles

- (2017) Nazarov A.I., Neterebskii B.O. **The Multiplicity of Positive Solutions to A Quasilinear Equation Generated By The Il'in–Caffarelli–Cohn–Nirenberg Inequality.** J Math Sci 224, 448–455. <https://doi.org/10.1007/s10958-017-3427-z>
- (2018) Yakovlev Pavel, Anikin Anton, Bolshakova Olga, Gasnikov Alexander, Gornov Alexander, Ermak Timofei, Makarenko Dmitrii, Morozov Vladimir, Neterebskii Bogdan. **Algorithms for local minimization of 3D molecules OPLS force field.** <https://arxiv.org/abs/1810.03358>
- (2022) Irina Isakova-Sivak, Ekaterina Stepanova, Victoria Matyushenko, Sergei Niskanen, Daria Mezhenkaya, Ekaterina Bazhenova, Elena Krutikova, Tatiana Kotomina, Polina Prokopenko, Bogdan Neterebskii, Aleksandr Doronin, Elena Vinogradova, Kirill Yakovlev, Konstantin Sivak, and Larisa Rudenko. **Development of a T Cell-Based COVID-19 Vaccine Using a Live Attenuated Influenza Vaccine Viral Vector.** <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9318028/>
- (2024) Ekaterina Stepanova, Irina Isakova-Sivak, Daria Mezhenkaya, Sergei Niskanen, Victoria Matyushenko, Ekaterina Bazhenova, Alexandra Rak, Pei Fong Wong, Polina Prokopenko, Tatiana Kotomina, Elena Krutikova, Sergei Legotskiy, Bogdan Neterebskii, Tatiana Ostroukhova, Konstantin Sivak, Yana Orshanskaya, Kirill Yakovlev, Larisa Rudenko. **Expression of the SARS-CoV-2 receptor-binding domain by live attenuated influenza vaccine virus as a strategy for**

**designing a bivalent vaccine against COVID-19 and influenza.** <https://pmc.ncbi.nlm.nih.gov/articles/PMC11003101/>

## Patents

- Influenza virus-based isolated recombinant virus ([WO2023287324A1](#))